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Started at	Sunday, April 24, 2022, 2:25 p.m
status	Completed
Finished at	Sunday, April 24, 2022, 2:40 p.m
Spent time	14 minutes 21 seconds
feedback	Congratulations, excellent performance (90-100%).

question **1**
Complete
Achievable points: 1.00

Assume a node in DCF mode with RTS/CTS activated wants to transmit multiple segments of data: are multiple RTS/CTS cycles required for this?

Choose one or more answers:

- [ ] a. Transmission of multiple segments of data is not possible with RTS/CTS activated
- [ ] b. Multiple RTS/CTS cycles are required in case collisions happen
- [ ] c. Yes, an RTS/CTS cycle is required for each segment
- [x] i.e. No, typically only a single RTS/CTS cycle is required

question **2**

Complete

Achievable points: 1.00

Consider the case that a new packet arrives internally from higher layers. For which cases does the backoff apply?

Choose one or more answers:

- [x] a. In the case that the NAV has indicated a busy medium
- [x] b. In the case that the medium is sensed as busy
- [ ] c. In the case that a collision is detected by another terminal

question **3**

Complete

Achievable points: 1.00

Exponential backoff: what is its purpose?

Choose one or more answers:

- [x] a. It delays retransmissions after collisions to reduce the risk of subsequent collisions
- [x] b. It reduces the risk of collisions compared to ALOHA
- [ ] c. It outperforms reservation-based schemes in throughput
- [x] d. It translates high network load in delay to avoid collisions
- [x] e. It tries to maximize the handled network load

Frage **4**
Vollständig
Erreichbare Punkte: 1,00

How is the exponential backoff time computed?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. It is drawn from a uniformly distributed pseudorandom waiting time from the interval [0, CW]
- [ ] b. It draws a nonuniformly distributed pseudorandom number from the interval [0, CW] multiplied by the slot time
- [x] c. In case of a repeated collision, the stations double the backoff time
- [x] d. It is computed from a uniformly distributed pseudorandom number from the interval [0, CW] multiplied by the slot time

Frage **5**
Vollständig
Erreichbare Punkte: 1,00

What is correct for DCF?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. DCF provides support for time-bounded data services
- [ ] b. DCF utilizes a contention-free MAC mechanism
- [x] c. DCF utilizes a contention-based MAC mechanism

Frage **6**
Vollständig
Erreichbare Punkte: 1,00

What is correct for DIFS?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Highest priority: for ACK, CTS, polling responses
- [ ] b. Medium priority for time-bounded service using PCF
- [x] c. Lowest priority for random access

Frage 7
Vollständig
Erreichbare Punkte: 1,00

What is correct for PCF?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. PCF utilizes a contention-free MAC mechanism
- [x] b. PCF provides some support for time-bounded data services
- [ ] c. PCF utilizes a contention-based MAC mechanism

Frage 8
Vollständig
Erreichbare Punkte: 1,00

What is correct for PIFS?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Lowest priority for random access
- [ ] b. Highest priority: for ACK, CTS, polling responses
- [x] c. Priority for polled stations over PCF

Frage 9
Vollständig
Erreichbare Punkte: 1,00

What is correct for SIFS?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Lowest priority: for random access
- [x] b. Highest priority: for ACK, CTS, polling responses
- [ ] c. Medium priority: for time-bounded service using PCF

Frage **10**
Vollständig
Erreichbare Punkte: 1,00

What is the purpose of different interframe spaces?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. To provide guaranteed QoS
- [x] b. Ensure the non-delayed transmission of ACKs
- [x] c. Differentiate the access priority between different types of messages
- [x] d. Ensure the non-delayed transmission of CTSs
- [ ] e. Ensure the non-delayed transmission of RTSs

Frage **11**
Vollständig
Erreichbare Punkte: 1,00

What is the purpose of the NAV?

Wählen Sie eine oder mehrere Antworten:

- [x] a. It tries to solve the inconsistency problem in wireless systems existing between different stations
- [x] b. NAV reduces the number of collisions
- [x] c. NAV indicates the duration of the reservation of the medium
- [ ] d. NAV indicates the priorities of stations

Frage **12**
Vollständig
Erreichbare Punkte: 1,00

Which mechanism is preferable for high-load situations?

Wählen Sie eine oder mehrere Antworten:

- [x] a. PCF ensures a better utilization of the medium
- [ ] b. DCF ensures a better utilization of the medium

Vorhergehende Aktivität

◀ Quiz: IEEE 802.11 - Basics

Direkt zu:

next activity

Quiz: IEEE 802.11 - MAC Management ▶

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